

Quick Summary Notes - Probability Distributions

Bernoulli Distribution

Used For	Used for single trial experiments with only two outcomes.
Example	Pass/Fail, Yes/No, Spam/Not Spam
Formula	$P(X=x)=p^x(1-p)^{(1-x)}$
Memory Trick	Checks whether an event happened or not.

Binomial Distribution

Used For	Used to count number of successes in fixed trials.
Example	Number of heads in 10 coin tosses
Formula	$P(X=x)=C(n,x)p^x(1-p)^{(n-x)}$
Memory Trick	Counts how many times success occurs.

Poisson Distribution

Used For	Used to count events occurring in a fixed interval.
Example	Calls per hour, customers per minute
Formula	$P(X=x)=e^{-\lambda} \lambda^x / x!$
Memory Trick	Counts how many times an event happens naturally over time.

Normal (Gaussian) Distribution

Used For	Used for continuous symmetric data around mean.
Example	Heights, exam marks, IQ scores
Formula	Bell-shaped distribution with mean μ and SD σ
Memory Trick	Most values stay near average.

Standard Normal Distribution

Used For	Used to standardize data for comparison.
Example	Z-score calculations
Formula	$z = (x - \mu) / \sigma$
Memory Trick	Shows how far a value is from mean.

Uniform Distribution

Used For	Used when all values have equal probability.
Example	Fair dice, random number generation
Formula	$f(x) = 1 / (b - a)$
Memory Trick	Every value has equal chance.

Log-Normal Distribution

Used For	Used for positively skewed continuous data.
Example	Income, stock prices, house prices
Formula	$\log(X)$ follows normal distribution
Memory Trick	Few very large values with many small values.

Power Law Distribution

Used For	Used when few large events dominate many small ones.
Example	Social media followers, YouTube views
Formula	$P(X=x) \propto x^{-\alpha}$
Memory Trick	Few huge events dominate the system.

Pareto Distribution

Used For	Used for 80/20 rule type problems.
Example	20% customers generate 80% revenue
Formula	Heavy-tail Pareto model
Memory Trick	Small percentage causes most impact.